Tuesday, October 6th, 2026

MIT Problem Set 0

0-4) Prove by induction

$$\sum_{i=1}^{n} i^3 = \left[\frac{n(n+1)}{2}\right]^2 \quad \text{for any integer } n \geq 1$$

Base case $n=1$, assume $n=k$ is true, prove $n=k+1$ is true

$$\sum_{i=1}^{1} i^3 = 1 = \left[\frac{1(1+1)}{2}\right]^2$$

$$= \left[\frac{2}{2}\right]^2$$

$$1 = 1 \quad \checkmark$$

$$\sum_{i=1}^{k} i^3 = \left[\frac{k(k+1)}{2}\right]^2$$

$$\sum_{i=1}^{k} i^3 + (k+1) = \left[\frac{k(k+1)}{2}\right]^2 + k+1$$